# Finite terms in the limit $\Lambda \to 0$, $\vee \to \infty$: check, simplification, and comparison with forward rapidity (AABKL, arXiv:1411.2869)

Input: your `Limiting_cases.tex` (in `input/`). Reference: the checked exact-$\vee$ remaining terms, `neglog/corrected/Other_terms.tex`. Notation: $a \equiv \frac{k^+}{\vee} \to 0$, $\alpha_s = \frac{g^2}{4\pi}$, $\beta_0 = \frac{11}{3}N_c$, $c_0 = 2e^{-\gamma_E}$, $\ell_\Delta(\mu) = \log\frac{(y'-y)^2\mu^2}{c_0^2}$. $d_\perp = 2-\epsilon$; in a finite term $d_\perp \to 2$. Blue = changed by me.

## 1 Answer

1. **Yes, these are the finite terms** of single inclusive gluon production at NLO at mid rapidity, after JIMWLK ($\log\frac{k^+}{\Lambda}$) and BFKL ($\log\frac{\vee}{k^+}$), with three provisos:
   - every equation is the correct $a \to 0$, $\Lambda \to 0$ limit of the checked exact-$\vee$ equation (Table 1: the difference falls exactly like $a$, to $10^{-9}$ at $a = 10^{-8}$);
   - Group II Row IV is not itself finite: it carries the DGLAP pole and the running coupling; its finite part is Eq. (3.10);
   - three small edits are needed (Sec. 2.1).
2. **The limits exist.** Group III, $\int_0^{\vee-k^+} dp^+ \to \int_0^\infty$: the integrands fall like $1/p^{+2}$ (Sec. 2.2). Group II Row IV, $\int_{k^+/\vee}^1 \to \int_0^1$: convergent because lines 1 and 2 are under one $\xi$ integral.
3. **Simplified** (Sec. 3):
   - c.c. $= 2\,\mathrm{Re}$ of the expectation value (Groups I, III);
   - every longitudinal integral becomes $\frac{1}{k^+}\times$ a dimensionless integral over $[0,1]$, so **the finite terms depend on $k^+$ only through the overall $\frac{1}{k^+}$ of LO**: they are rapidity independent;
   - Group I Row V: first part in closed form (the entire function Ein); $2a$ and $2b$ have the same kernel and combine into one $\xi$ integral;
   - Group III: with $\zeta = \frac{k^+}{p^+ + k^+}$ the measures are $\frac{d\zeta}{\zeta}$, $\left[\frac{1}{1-\zeta}\right]_+ d\zeta$, $\frac{d\zeta}{\zeta^2}$ and the phase is $e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}$; one line in closed form;
   - Group I Row II part 1 + c.c. $= \frac{\alpha_s N_c}{2\pi}\left(\frac{67}{9}-\frac{\pi^2}{3}\right)\mathrm{LO} = \frac{\alpha_s}{2\pi}2K\,\mathrm{LO}$ ($K$: the CMW constant).

   Final result: Eq. (3.16).
4. **AABKL** (Sec. 4): the virtual kernel, the real kernel, the limits ($\vee \leftrightarrow x_p$), the shifted transverse coordinates, the fragmentation-function DGLAP and the scheme constant are the same. Different, and expected to be: the CMW constant (our LO $\propto \alpha_s$), the terms of a projectile made of colour charges (three-$\rho$ rows, emission before the target from the charges, linear polarization, $H_4$), and the scheme of the soft subtraction.

## 2 Check of the limits

Analytic limits (`checks.py`): $R^i$: $\log\frac{p^2}{k^2+ap^2} \to \log\frac{p^2}{k^2}$, the $a$ and $\log(1-a)$ terms $\to 0$; Group II Row IV: $\int_{k^+/\vee}^1 \to \int_0^1$, $(1-a)H_4 \to H_4$, the $\log(1-a)$ line is absorbed in the plus distribution and vanishes.

### 2.1 Edits needed

E1 **Group I Row II part 1**: $+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]$ must go: $\log(1-a)\log\frac{1}{a} \to -a\log\frac{1}{a} \to 0$ ($-9.2\times10^{-4}$ at $a = 10^{-4}$, $-1.8\times10^{-7}$ at $10^{-8}$). The bracket is $\left\{-\frac{67}{9}+\frac{\pi^2}{3}\right\}$.

E2 **Group I Row V,** $d^3N_{2b\,\mathrm{finite}}$, third line: $\int_{k^+-\Lambda}^{\Lambda} \to \int_{k^+}^0$ (the integrand is regular at both ends; the line above already has $\int_{k^+}^0$).

E3 **Group III**: $\int_0^{\vee-k^+} \to \int_0^\infty$ in the limit (allowed: Sec. 2.2).

Also: $d_\perp \to 2$ in the finite rows (G1 V $2a$, $2b$; G2 I $2f$; G3 III part 2), where you still have $d_\perp$.

Table 1: $\max|\text{exact}(a)-\text{yours}|/\max|\text{yours}|$ for the equation as a whole, at $a=10^{-2},10^{-4},10^{-6},10^{-8}$ (three field/point configurations; smooth SU(3) Wilson lines and charges; $p^+=0.37k^+$; `limcheck.py`). $\propto a$ (or $a^2$) everywhere: every equation is the limit.

| equation | configuration 1 (2 and 3 alike, `limcheck.json`) |
|---|---|
| G1 III parts 1, 2 | $7.4\times10^{-3},\ 7.4\times10^{-5},\ 7.4\times10^{-7},\ 7.4\times10^{-9};\quad 2.0\times10^{-3},\dots,2.0\times10^{-9}$ |
| G1 III 3a, 3b; G1 V (all) | 0 (unchanged) |
| G1 IV; G2 I 1a; G3 I, II | $3.6\times10^{-3},\dots,3.6\times10^{-9}$ |
| G2 I 1b, 2a, 2c, 2e | $1.9\times10^{-2},\dots,1.9\times10^{-8}$ etc. ($\propto a$) |
| G2 I 2b, 2d, 2f | $4\times10^{-5},\ 4\times10^{-9},\ 4\times10^{-13},\ 2\times10^{-16}$ ($\propto a^2$) |
| G3 III parts 1, 2; V; VI | $7.4\times10^{-4},\dots,7.4\times10^{-10}$ etc. ($\propto a$) |

### 2.2 Finiteness

- **Transverse (UV).** Taking $a\to0$ does not change the short-distance behaviour, so the result of `uv_check/UV_divergences.pdf` holds. Finite, provided these are kept as one expression: Group I Row III (parts 1, 2, 3a, 3b together), Group I Row V $2b$ finite (with its $\frac{11}{6}$ line), Group II Row I $2f$ (three lines). Group I Row II $d^3N_{2b}$ ($R^i$) is UV finite; its $p\to0$ and $k-p\to0$ ends (large $|x-y|$) are the IR point F5 of that note, unchanged by the limit.
- **Group III at large $p^+$.** The $y'$ integral is a Fourier transform at $q=k\big(1+\frac{p^+}{k^+}\big)$ of $\frac{(y'-z)^m}{(y'-z)^2}g(y')$, with $g$ smooth once $x'$ is integrated. Line 1 has the weights $\frac{1}{p^+{+}k^+}$, $\big[\frac{k^+}{p^+(p^+{+}k^+)}\big]_+\sim\frac{1}{p^{+2}}$, and the transform is $O(1/q)$: integrand $\sim\frac{1}{p^{+2}}$. Line 2 has weight $\frac{1}{k^+}$ but its bracket vanishes at $y'=z$ (the $(x'-z)$, $U(z)$ term), so its transform is $O(1/q^2)$: integrand $\sim\frac{1}{p^{+2}}$. Numerically (`largep.py`, model $g$): $q|I_{\rm unsub}|=4.876,\ 4.885,\ 4.891,\ 4.893,\ 4.893\to2\pi g(0)=4.893$ and $q^2|I_{\rm sub}|=5.36,\ 5.40,\ 5.39,\ 5.38,\ 5.39$ at $q=5,\dots,80$. The limit $\vee\to\infty$ exists row by row.
- **Group II Row IV at $\xi\to0$.** $\frac{1-\xi}{\xi}\big(\mathrm{LO}_\xi-\mathrm{LO}_\xi^y\big)$ is integrable (the phase forces $y'\to y$); separately each would give $\log\frac{\vee}{k^+}$. Keep them under one integral, as you do.

## 3 Simplification

### 3.1 The c.c.

The charges are hermitian and the adjoint Wilson lines real, so the c.c. of a term is: $i\to-i$, $e^{-ik\cdot(\cdot)}\to e^{+ik\cdot(\cdot)}$, and the order of the $\rho$'s reversed. Its expectation value in the projectile is the complex conjugate. Hence, for Groups I and III,

$$\big\langle\, d^3N_{\rm row}+\text{c.c.}\,\big\rangle=2\,\mathrm{Re}\,\big\langle d^3N_{\rm row}\big\rangle,\qquad\text{and for the rows with } i\,f\text{:}\quad 2\,\mathrm{Re}\,\langle iX\rangle=-2\,\mathrm{Im}\,\langle X\rangle. \tag{3.1}$$

Group II is self-conjugate ($1a\leftrightarrow1b$; in part 2 every term comes with its $(x,w)\leftrightarrow(x',w')$ partner).

### 3.2 Group I

**Row II, part 1 (CMW constant).** Relative to LO the prefactor is $\frac{\alpha_sN_c}{4\pi}$, the c.c. doubles it, and $[U(x')-U(w')][U(w)-U(x)]=-[U(w')-U(x')][U(w)-U(x)]$:

$$d^3N_{\mathrm{G1\,II}(1)}+\text{c.c.}=\frac{\alpha_sN_c}{2\pi}\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\mathrm{LO}=\frac{\alpha_s}{2\pi}\,2K\,\mathrm{LO},\qquad K=N_c\Big(\frac{67}{18}-\frac{\pi^2}{6}\Big). \tag{3.2}$$

Since LO $\propto\alpha_s$, absorbing it into the coupling of LO needs $\alpha_s\to\alpha_s\big(1+\frac{\alpha_s}{\pi}K\big)$, twice the CMW shift; keep it as an explicit constant.

**Row II, part 2 ($R^i$), Rows III, IV:** as in your file (no longitudinal integral left), with E1 and +c.c.

**Row V, first part: one $\xi$ integral and closed form.** With $\xi=\frac{p^+}{k^+}$ in line 1 and $\xi=1-\frac{q^+}{k^+}$ in line 2 both phases become $e^{-ik\cdot(w'-(1-\xi)z-\xi x)}$, and $\frac{1}{k^+}\int_0^{k^+}dq^+\big[\frac{1}{q^+}\big]_+=\int_0^1d\xi\big[\frac{1}{1-\xi}\big]_+$:

$$\begin{aligned}\frac{d^3N_{\mathrm{V}(1)}}{d^3k}&=\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_0^1d\xi\int_{x,z,y,x',y',w'}e^{-ik\cdot(w'-(1-\xi)z-\xi x)}\frac{(x'-w')^i(z-x)^m(y-z)^k(y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2}\\&\quad\times\Big[\delta_{im}\delta_{jk}-\Big[\frac{1}{\xi}\Big]_+\delta_{jm}\delta_{ik}-\Big[\frac{1}{1-\xi}\Big]_+\delta_{ij}\delta_{mk}\Big]\,\mathcal{C}_{\mathrm{V}(1)}\qquad(3.3)\\&=\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_{x,z,y,x',y',w'}e^{-ik\cdot(w'-z)}\frac{(x'-w')^i(z-x)^m(y-z)^k(y'-x)^j}{(x'-w')^2(z-x)^2(y-z)^2(y'-x)^2}\\&\quad\times\Big[\frac{1-e^{-i\alpha}}{i\alpha}\,\delta_{im}\delta_{jk}+\mathrm{Ein}(i\alpha)\,\delta_{jm}\delta_{ik}+e^{-i\alpha}\mathrm{Ein}(-i\alpha)\,\delta_{ij}\delta_{mk}\Big]\,\mathcal{C}_{\mathrm{V}(1)},\qquad\alpha\equiv k\cdot(z-x),\qquad(3.4)\end{aligned}$$

$\mathrm{Ein}(s) = \int_0^1 \frac{1-e^{-st}}{t}dt = \sum_{n\geq 1}\frac{(-1)^{n+1}s^n}{n\,n!} = \gamma_E + \log s + E_1(s)$ (entire), and $\mathcal{C}_{\mathrm{V}(1)} = \big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\big[f^{adc}U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\big]$.

**Row V, second part: $2a$ and $2b$ finite combine.** With $\xi = \frac{p^+}{k^+}$ in the $p^+$ lines and $\xi = 1 - \frac{q^+}{k^+}$ in the $q^+$ lines, all lines of $2a$ and of $2b$ finite have the same phase, the same denominator $k^+ D_\xi$,

$$D_\xi \equiv \xi\,(y-x)^2 + (1-\xi)\,(y-z)^2, \tag{3.5}$$

and the same kernel, Eq. (3.7); they differ only by the colour factor. Therefore (E2 used, $d_\perp \to 2$):

$$\begin{aligned}\frac{d^3N_{2a}}{d^3k} + \frac{d^3N_{2b\,\mathrm{fin}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_0^1 d\xi \int_{x,y,z,x',w'} \frac{e^{-ik\cdot(w'-\xi x-(1-\xi)z)}}{D_\xi}\,\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\,\mathcal{B}^{im}(\xi)\\ &\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_cU^{af}(y) - f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)\Big]\rho^{b'}(x')\rho^f(y)\\ &\quad+\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\frac{11}{6}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\\ &\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_cU^{af}(y) - N_cU^{af}(z)\Big]\rho^{b'}(x')\rho^f(y),\end{aligned} \tag{3.6}$$

$$\begin{aligned}\mathcal{B}^{im}(\xi) &= \xi(1-\xi)\,\frac{\delta^{im}\big[(y-x)^2-(y-z)^2\big]}{(x-z)^2} + (1-\xi)\,\delta^{im}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\ &\quad+\xi\,\delta^{im}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big] - \Big(\Big[\frac{1}{\xi}\Big]_+ - 1\Big)S^{im} - \Big(\Big[\frac{1}{1-\xi}\Big]_+ - 1\Big)T^{im} - S'^{im},\end{aligned} \tag{3.7}$$

$$S^{im} = \frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}, \qquad T^{im} = \frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2},$$
$$S'^{im} = \frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}.$$

The $2a$ colour factor vanishes at $z = x$ ($-f^{adc}f^{fbe}U^{db}(x)U^{ce}(x) + N_cU^{af}(x) = 0$, checked), so the $x \to z$ behaviour of the sum is that of $2b$, removed by the $\frac{11}{6}$ line: Eq. (3.6) is finite as written. The weights $\xi(1-\xi)$, $1-\xi$, $\xi$, $\big[\frac{1}{\xi}\big]_+$, $\big[\frac{1}{1-\xi}\big]_+$ are those of the virtual $g \to gg$ kernel, and $\int_0^1 d\xi\,[\xi(1-\xi)-2] = -\frac{11}{6}$ (Sec. 4).

## 3.3 Group II

**Row I:** as in your file (no c.c.; no longitudinal integral; $d_\perp \to 2$ in $2f$, three lines together).

**Row IV: DGLAP + running coupling + finite.** Your equation, with $\mathcal{P}_{gg} = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big] + \frac{\beta_0}{2}\delta(1-\xi)$ on $[0,1]$ and the brackets inside $\frac{d^3N_{\xi,\mathrm{LO}}}{d^3k}$ (`g2iv_rc/compact`), gives

$$\text{DGLAP:}\quad -\frac{\alpha_s}{2\pi}\int_0^1\frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - 2N_c\frac{1-\xi}{\xi}\mathrm{LO}^y_\xi\big[\ell_\Delta(\mu_F)\big]\Big\}\quad(\text{pole in } D_{h/g}(\mu_F)), \tag{3.8}$$

$$\text{running coupling:}\quad \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\ell_\Delta(\mu_R)\big] \quad\Rightarrow\quad \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)}, \tag{3.9}$$

$$\text{finite:}\quad -H_4 + \frac{\alpha_s}{2\pi}\,2N_c\int_0^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}^{\mathrm{lp}}_\xi\Big|_{d_\perp=2}. \tag{3.10}$$

## 3.4 Group III: $\zeta = \frac{k^+}{p^+ + k^+}$

With $p^+ = k^+\frac{1-\zeta}{\zeta}$ ($p^+ : 0 \to \infty$ is $\zeta : 1 \to 0$; E3 used):

$$\frac{dp^+}{p^+ + k^+} = \frac{d\zeta}{\zeta}, \qquad \Big[\frac{k^+}{p^+(p^+ + k^+)}\Big]_+ dp^+ = \Big[\frac{1}{1-\zeta}\Big]_+ d\zeta, \qquad \frac{dp^+}{k^+} = \frac{d\zeta}{\zeta^2}, \tag{3.11}$$

$$e^{-ik\cdot(y'-w)}e^{-ik\cdot\frac{p^+}{k^+}(y'-z)} = e^{-i\frac{k}{\zeta}\cdot\left(y'-\zeta w-(1-\zeta)z\right)} \tag{3.12}$$

(your plus prescription, subtraction at $p^+ = 0$, is the standard one at $\zeta = 1$; all checked numerically). Rows III part 1, V, VI then read, with $P_{\mathrm{III}(1)} = \frac{ig^4 f^{c'd'a}}{16\pi^5}$, $P_{\mathrm{V}} = \frac{ig^4 f^{c'ca}}{8\pi^5}$, $P_{\mathrm{VI}} = \frac{ig^4 f^{c'd'a}}{8\pi^5}$,

$$\begin{aligned}\frac{d^3N_{\mathrm{G3\,X}}}{d^3k} &= \frac{1}{(2\pi)^3}\,P_{\mathrm{X}}\,\frac{1}{k^+}\int_0^1 d\zeta\int e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\Big\{\Big[\frac{1}{\zeta}\,(y'-z)\cdot(x'-y')\,(y-w)\cdot(x-z)\\ &\quad-\Big(\Big[\frac{1}{1-\zeta}\Big]_+ + \frac{1}{\zeta}\Big)(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)\Big]\frac{\mathcal{C}_{\mathrm{X}}\,\mathcal{C}'_{\mathrm{X}}}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\end{aligned}$$

$$- \frac{1}{\zeta^2}\,\frac{(y'-z)\cdot(y-w)}{(y'-z)^2(y-w)^2(x-z)^2}\Big[\frac{(x-z)\cdot(x'-y')}{(x'-y')^2}\,\mathcal{C}_{\rm X} - \frac{(x-z)\cdot(x'-z)}{(x'-z)^2}\,\mathcal{C}_{\rm X}\big|_{U(y')\to U(z)}\Big]\mathcal{C}'_{\rm X}\Big\}, \tag{3.13}$$

with the colour factors of your file: $\mathcal{C}_{\rm III(1)} = \mathcal{C}_{\rm VI} = \big[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')\big]\rho^{e'}(x')$, $\mathcal{C}_{\rm V} = \big[U^{e'c'}(y') - U^{c'e'}(x')\big]\rho^{e'}(x')$; $\mathcal{C}'_{\rm III(1)} = U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\}$, $\mathcal{C}'_{\rm V} = \big[U^{ce}(x)U^{ad}(y) - U^{ad}(w)U^{ce}(x)\big]\rho^e(x)\rho^d(y)$, $\mathcal{C}'_{\rm VI} = U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - U^{ad}(y)\rho^d(y)\rho^b(x)$. The second line is the $y'\to z$ subtraction; it is what makes $\zeta\to 0$ ($p^+\to\infty$) integrable.

**Row III, part 2.** With $A\equiv(x-z)^2$, $B\equiv(x-w)^2$, $D_\zeta\equiv(1-\zeta)A+\zeta B$, $\frac{dp^+}{p^+A+k^+B} = \frac{d\zeta}{\zeta D_\zeta}$:

$$\begin{aligned}
\frac{d^3N_{\rm III(2)}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_0^1 d\zeta \int_{w,x',y',z,x} e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\,\frac{(y'-z)^m(x'-y')^{k'}}{(y'-z)^2(x'-y')^2}\Big\{\frac{\delta_{k'm}}{D_\zeta}\Big[\frac{(1-\zeta)(A-B)}{(z-w)^2}+\frac{1-\zeta}{\zeta}\,b+c\Big]\\
&\quad - \frac{1}{\zeta D_\zeta}\,e^{k'm} - \Big(\Big[\frac{1}{1-\zeta}\Big]_+\frac{\zeta}{D_\zeta}+\frac{1}{D_\zeta}\Big)d^{k'm}\Big\}\,\mathcal{C}_{\rm VI}\,\mathcal{C}''\\
&\quad - \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_0^1 d\zeta\,\frac{1-\zeta}{\zeta^2}\int e^{-i\frac{k}{\zeta}\cdot(y'-\zeta w-(1-\zeta)z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{t^{k'm}}{D_\zeta}\Big[\frac{(x'-y')^{k'}}{(x'-y')^2}\mathcal{C}_{\rm VI} - \frac{(x'-z)^{k'}}{(x'-z)^2}\mathcal{C}_{\rm VI}\big|_{U(y')\to U(z)}\Big]\mathcal{C}''\\
&\quad + \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m(x'-z)^{k'}}{(w'-z)^2(x'-z)^2}\,\frac{(x-w)^2\log\frac{(x-w)^2}{(x-z)^2}}{(x-z)^2\big[(x-w)^2-(x-z)^2\big]}\,t^{k'm}\,\mathcal{C}_{\rm VI}\big|_{U(y')\to U(z)}\,\mathcal{C}'',
\end{aligned} \tag{3.14}$$

where $\mathcal{C}'' = f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$ (times $f^{c'd'a}$ in $\mathcal{C}_{\rm VI}$), and $b, c, d^{k'm}, e^{k'm}, t^{k'm}$ are your brackets: $b = \frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2(x-w)^2} - \frac{A-B}{2(z-w)^2}$, $c = \frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2(x-z)^2} - \frac{A-B}{2(z-w)^2}$, $d^{k'm} = \frac{(x-w)^{k'}(z-w)^m}{(z-w)^2} + \frac{(x-z)^m(x-w)^{k'}}{2(x-z)^2}$, $e^{k'm} = \frac{(x-z)^m(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m(x-w)^{k'}}{2(x-w)^2} + \frac{(x-w)^m(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}(x-w)^m}{2(x-z)^2}$, $t^{k'm} = \frac{(x-z)^{k'}(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}(x-w)^m}{2(x-w)^2}$. A plus distribution acts on everything to its right, phase included ($\int_0^1 d\zeta\big[\frac{1}{1-\zeta}\big]_+\Phi(\zeta) = \int_0^1 d\zeta\,\frac{\Phi(\zeta)-\Phi(1)}{1-\zeta}$). The last line is your third line with its $p^+$ integral done:

$$\int_0^\infty dp^+\,\frac{B}{A\,(p^++k^+)(p^+A+k^+B)} = \frac{B\log\frac{B}{A}}{k^+A(B-A)} \qquad \text{(regular at } A=B\text{; checked to } 10^{-15}\text{)}. \tag{3.15}$$

### 3.5 Final result

$$\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_0^1\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s\to\alpha_s(c_0/|y'-y|)} + \frac{\alpha_sN_c}{2\pi}\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\mathrm{LO} + \log\frac{k^+}{\Lambda}\,(\mathrm{JIMWLK}\otimes\mathrm{LO}) + \log\frac{\vee}{k^+}\,(\mathrm{BFKL}\otimes\mathrm{LO}) + (3.8)\\
&\quad - H_4 + \frac{\alpha_s}{2\pi}2N_c\int_0^1 d\xi\,\frac{1-\xi}{\xi}\mathrm{LO}^{\rm lp}_\xi + \sum_{\rm G2\,I}\frac{d^3N}{d^3k}\\
&\quad + 2\mathrm{Re}\Big[\frac{d^3N_{\rm G1\,II(2b)}}{d^3k} + \frac{d^3N_{\rm G1\,III}}{d^3k} + \frac{d^3N_{\rm G1\,IV}}{d^3k} + (3.4) + (3.6) + \sum_{\rm X=III(1),V,VI}(3.13) + (3.14) + \frac{d^3N_{\rm G3\,I}}{d^3k} + \frac{d^3N_{\rm G3\,II}}{d^3k}\Big].
\end{aligned} \tag{3.16}$$

Every finite term is $\frac{1}{k^+}\times$ a function of $\mathbf{k}$ only. The constant $\frac{\alpha_s}{2\pi}2K$ LO is kept explicit (absorbing it would need twice the CMW shift).

# 4 Comparison with forward rapidity (AABKL, arXiv:1411.2869)

AABKL: a collinear parton of the proton (PDF $x_pf(x_p)$) at forward rapidity, LO $O(\alpha_s^0)$, final result Eq. (4.22) (gluon channel). Ours: a gluon radiated by the charges $\rho$ of a dilute projectile at mid rapidity, LO $O(\alpha_s)$. The dictionary is $x_p \leftrightarrow a = \frac{k^+}{\vee}$; $\vee\to\infty$ is $x_p\to 0$.

| AABKL | ours (this limit) | relation |
|---|---|---|
| virtual term: $\int_0^1 d\xi\big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{[1-\xi]_+}+\xi(1-\xi)\big] = -\frac{11}{6}$ | G1 V $2a+2b$, Eq. (3.6): weights $\xi(1-\xi)$, $1-\xi$, $\xi$, $[\frac{1}{\xi}]_+$, $[\frac{1}{1-\xi}]_+$ on $[0,1]$; G1 V(1), Eq. (3.3) | same kernel, same range; our $\frac{11}{6}$ line is their $-\frac{11}{6}$, i.e. $\beta_0 = 2C_A\frac{11}{6}$ |
| real term $\int_0^{1-x_p} d\xi\big[\frac{1-\xi}{[\xi]_+}+\frac{\xi}{1-\xi}+\xi(1-\xi)\big]$, $\xi$ = fraction of the unmeasured gluon | Group III, Eq. (3.13): $\int_a^1 d\zeta$ with $\big[\frac{1}{1-\zeta}\big]_+$ and $\frac{1}{\zeta}$; $1-\zeta$ = fraction of the unmeasured gluon | $\xi_{\rm AABKL} = 1-\zeta$; upper limit $1-x_p \leftrightarrow \zeta\ge a$ (our $\int_0^{\vee-k^+}dp^+$); $\to\int_0^1$ for $x_p, a\to 0$ |
| shifted dipoles $s\big[y,(1-\xi)\bar y+\xi z\big]$ | phase $e^{-i\frac{k}{\zeta}\cdot(y'-[\zeta w+(1-\zeta)z])}$ (G3); $e^{-ik\cdot(w'-[(1-\xi)z+\xi x])}$ (G1 V) | the same weighted centre of the parent; ours in Fourier form |

| | | |
|---|---|---|
| FF term: $P_{gg}$, $\log\big((y-\bar{y})^2\mu^2\big)$, scheme $f_{\mu^2} = f^{\overline{\rm MS}}_{1.26\mu^2}$ | G2 IV, Eq. (3.8), $\ell_\Delta = \log\frac{(y'-y)^2\mu^2}{c_0^2}$ | same $P_{gg}$; $R^2 = c_0^2 = 4e^{-2\gamma_E} = 1.26$ |
| soft term $\int_0^1 \frac{d\xi}{\xi}[A_\xi(y-z) - A_\xi(\bar{y}-z)]^2[\dots]$ (Ioffe-time cutoff; "not an extra evolution") | the subtraction lines: $\frac{1}{\zeta^2}[\dots]_{y'\to z}$ in Eq. (3.13), $\frac{1-\xi}{\xi}\mathrm{LO}^y_\xi$ in Eq. (3.8) | both are what is left of the soft region after the evolution is taken out; scheme dependent (Ioffe time vs. sharp $p^+$ cut), so not equal term by term |
| target evolution BK | $\log\frac{k^+}{\Lambda}$ JIMWLK⊗LO | same (BK = large-$N_c$ dipole form) |
| projectile: PDF DGLAP; no BFKL ($\ln\frac{1}{x_p} = O(1)$) | no PDF DGLAP (emission before the target from the charges is finite, G2 I); BFKL over $\log\frac{\vee}{k^+}$ | at mid rapidity $\log\frac{\vee}{k^+} = \frac{1}{2}\log\frac{\vee}{\Lambda}$ is large |
| no CMW constant (LO $O(\alpha_s^0)$) | $\frac{\alpha_s N_c}{2\pi}(\frac{67}{9} - \frac{\pi^2}{3})$LO | ours renormalizes the coupling of LO |
| – | three-$\rho$ rows (G1 II, III, IV, V(1); G3 I, II, III(1), V, VI), $H_4$, linear polarization | only for a projectile of colour charges; a single parton has none |
| quark channels ($C_F$) | none ($n_f = 0$) | – |

**Conclusion.** In the limit $\Lambda \to 0$, $\vee \to \infty$ our finite terms have AABKL's structure wherever the physics is the same: the virtual and real $g \to gg$ kernels with the same plus distributions on $[0,1]$, the same limits ($x_p \leftrightarrow \frac{k^+}{\vee}$), the same weighted transverse centres, the same fragmentation DGLAP in the same scheme. They differ where they must: our projectile is a set of colour charges (three-$\rho$, $H_4$, linear polarization, finite before-target emission), our LO carries $\alpha_s$ (running coupling and the CMW constant), and the soft remainders are scheme dependent. A numerical term-by-term equality is therefore not expected.

# 5 Files

`nlo_gluon_paper/limiting/`: `limcheck.py` (Table 1), `largep.py` (Sec. 2.2), `checks.py` (limits, CMW prefactor, $\zeta$ measures and plus prescription, phase identities, Ein closed forms, the closed-form $p^+$ integral, the $q^+ \to \xi$ map, the $2a$ colour at $z = x$), `limblocks.py`.